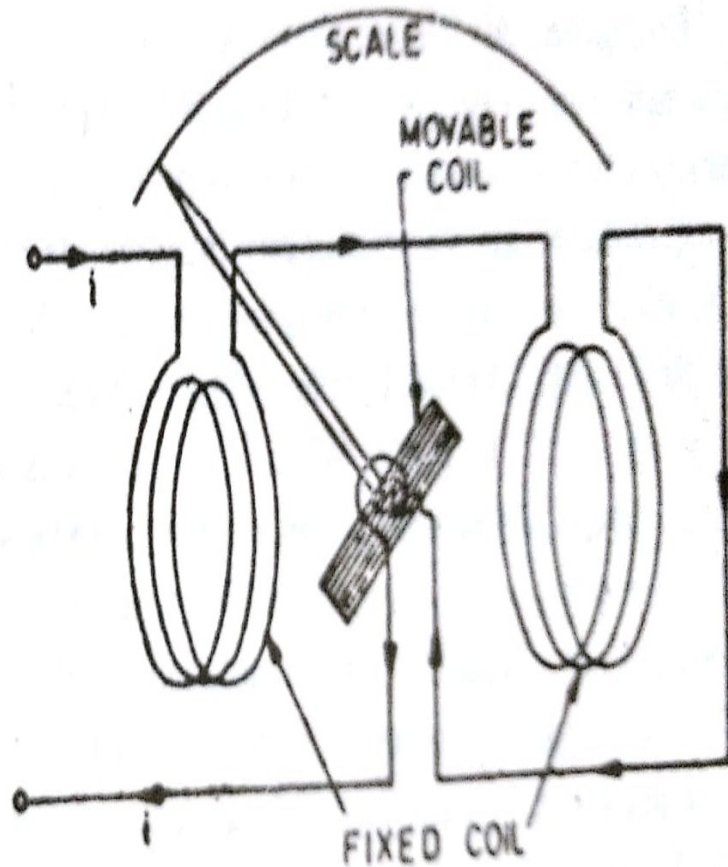


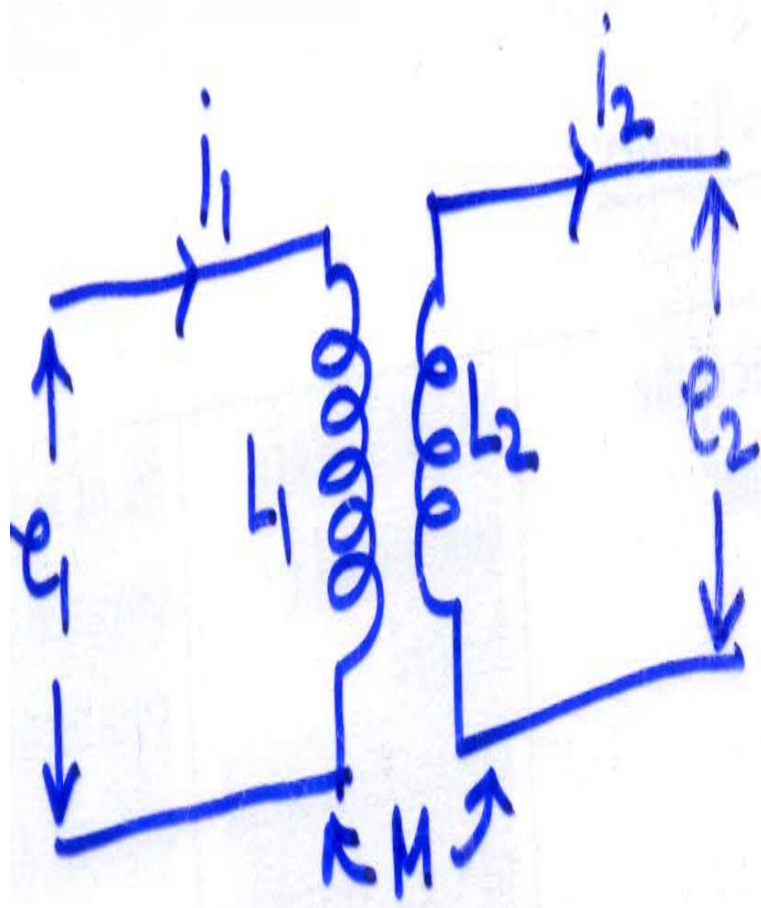
Power Measurement

Electrodynamometer Instrument



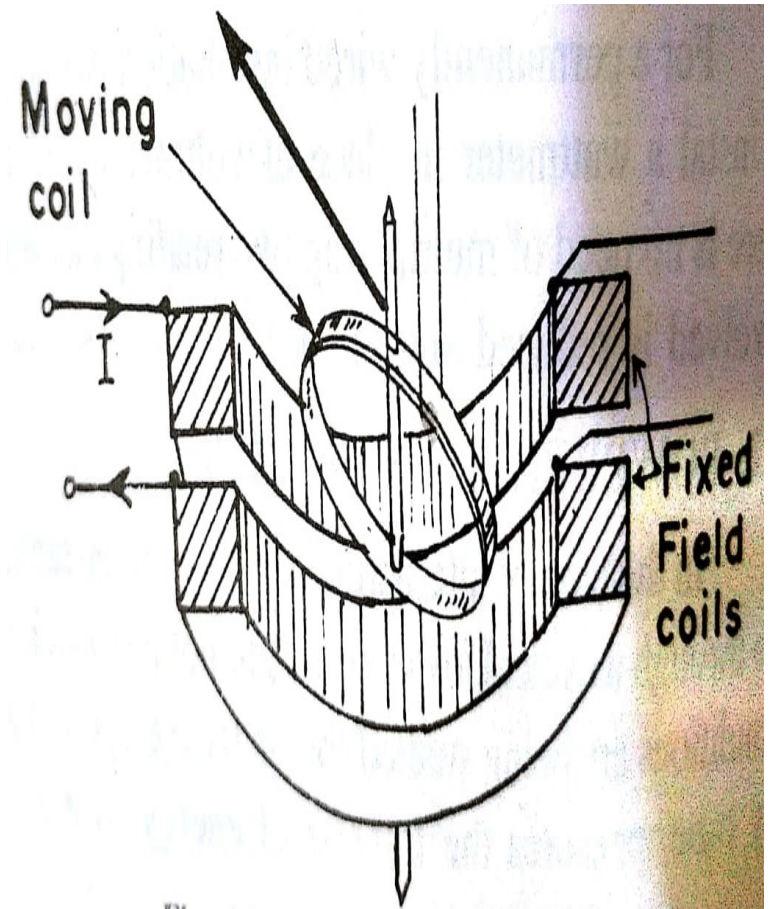
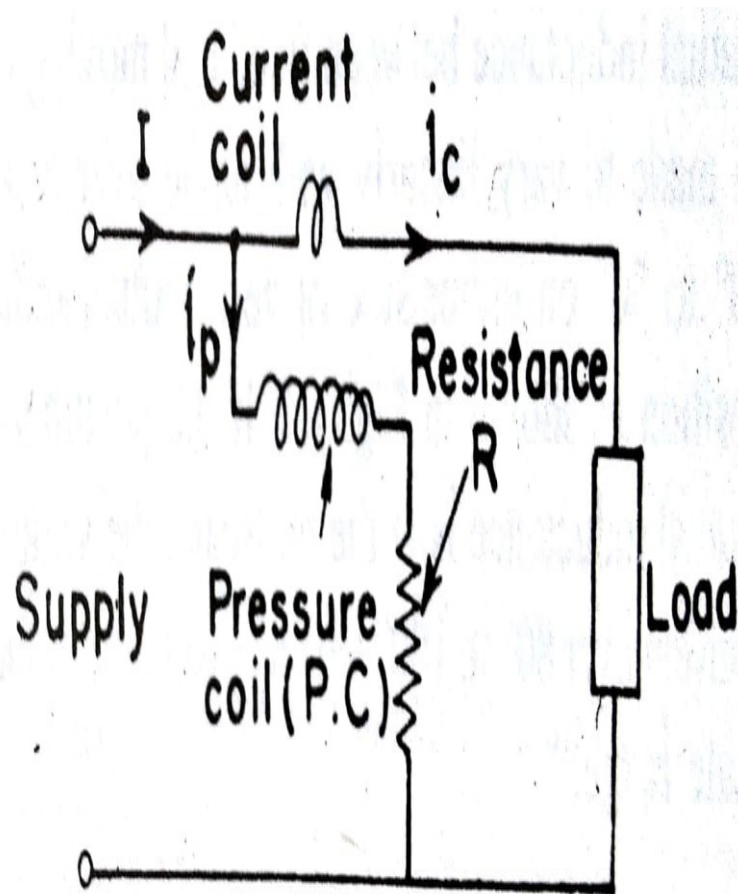
- Transfer Instrument- can be used in both ac and dc.
- Highly accurate.

Torque Equation of Electrodynamometer Instruments



$$T_i = i_1 i_2 \frac{dM}{d\theta}$$

Electrodynamometer Wattmeters



Construction

- Fixed Coils
- Moving Coil
- Spring Control
- Air friction damping
- Mirror type scales and knife edge pointer

Theory

$$\text{Instantaneous Torque } (T_i) = i_1 i_2 \frac{dM}{d\theta}$$

$$\text{Voltage across PC } (V) = \sqrt{2} V \sin \omega t$$

$$i_p = V/R_p = \sqrt{2} I_p \sin \omega t$$

$$i_c = \sqrt{2} I_c \sin (\omega t - \phi) \approx \sqrt{2} I \sin (\omega t - \phi)$$

$$\Rightarrow T_i = (\sqrt{2} I_p \sin \omega t) (\sqrt{2} I \sin \omega t - \phi) \frac{dM}{d\theta}$$

$$\Rightarrow T_i = I_p I [\cos \phi - \cos (2\omega t - \phi)] \frac{dM}{d\theta}$$

$$\text{Average deflecting torque } (T_d) = \frac{1}{2\pi} \int_0^{2\pi} T_i d(\omega t)$$

$$= I_p I \cos \phi \frac{dM}{d\theta}$$

$$= \frac{V}{R_p} I \cos \phi \frac{dM}{d\theta}$$

$$T_c = K\theta$$

At equilibrium

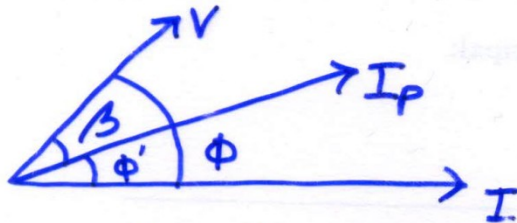
$$K\theta = \frac{V}{R_p} I \cos \phi \frac{dM}{d\theta}$$

$$\Rightarrow \theta = \left(\frac{1}{K R_p} \right) (V I \cos \phi) \left(\frac{dM}{d\theta} \right)$$

$$\Rightarrow \theta = K_1 P \frac{dM}{d\theta}$$

Errors

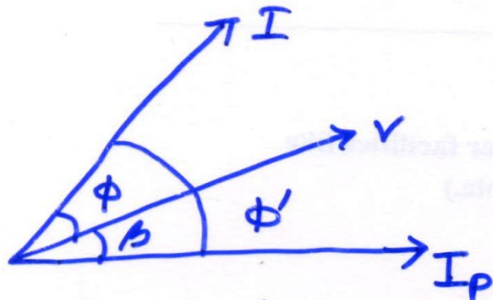
(1) Pressure Coil Inductance



Phasor for lagging p.f.

⇒ Wattmeter reads high for lagging loads

$$\text{Correction factor} = \frac{\cos \phi}{\cos \beta \cos(\phi - \beta)}$$



Phasor for leading p.f.

⇒ Wattmeter reads low for leading p.f. loads

$$\text{Correction factor} = \frac{\cos \phi}{\cos \beta \cos(\phi + \beta)}$$

$$\text{Error} = \tan \phi \tan \beta (V I \cos \phi) = V I \sin \phi \tan \beta$$

* Error compensated by using shunt capacitor.

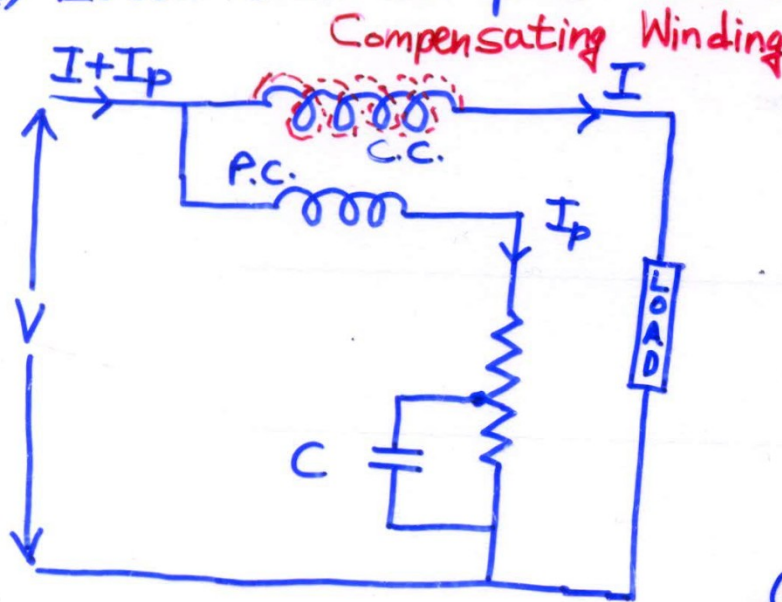
(2) Error due to mutual inductance effects.

(3) Errors caused because of connections.

LPF WATTMETERS

Ordinary wattmeter cannot be used $\therefore$

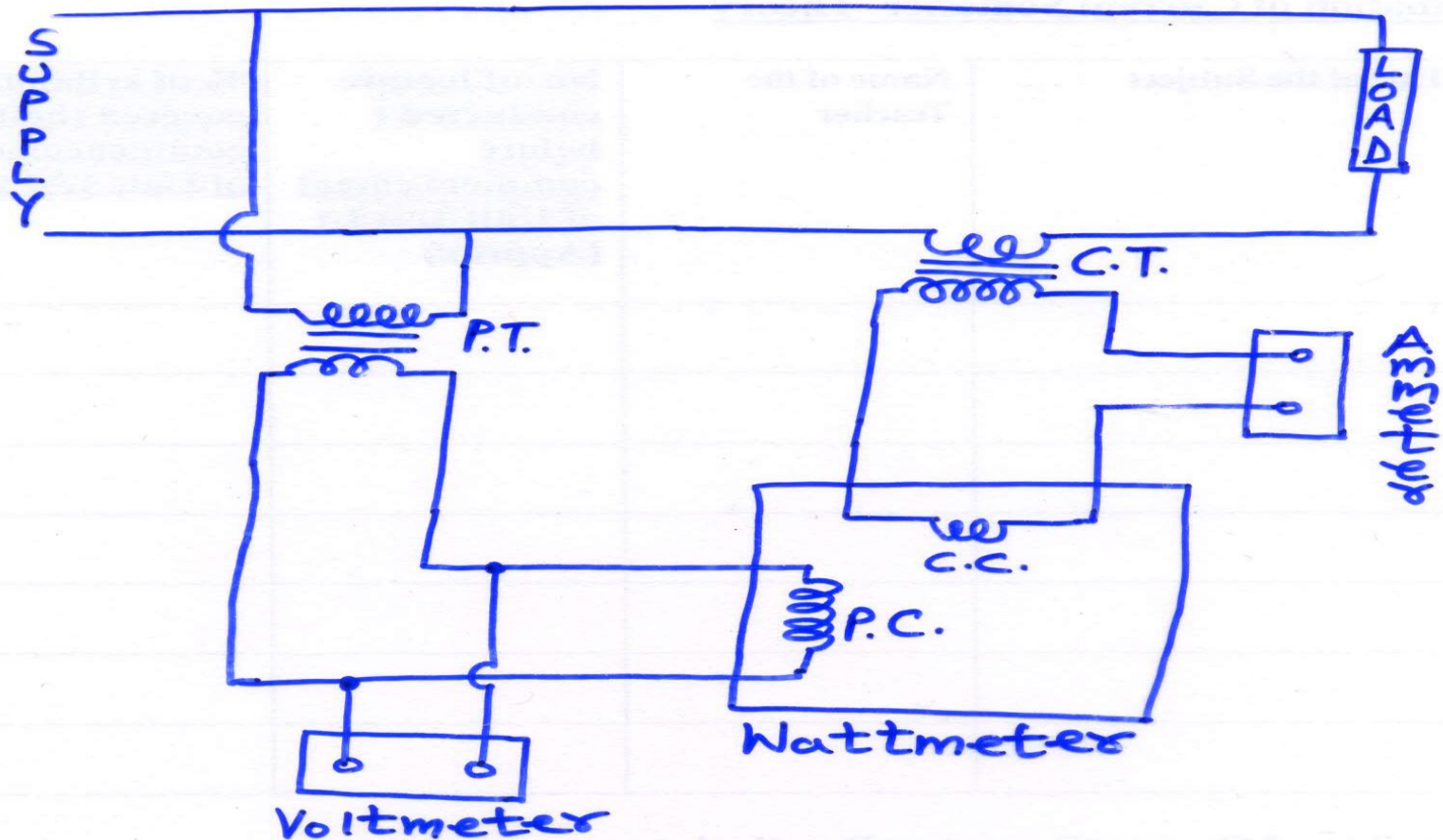
- (i) Small T_d
- (ii) Error due to p.c. inductance increases



FEATURES:

- (1) Pressure Coil Current
Almost ten times higher current as compared to normal wattmeter. This gives increased operating torque.
- (2) Compensation for inductance of p.c.
By connecting 'C'

Measurement of Power using Instrument Transformers



$$\text{Power} = \frac{\cos \phi}{\cos \beta \cos \alpha} \times (RCF)_{PT} \times (RCF)_{CT} \times (K_n)_{PT} \times (K_n)_{CT} \times W$$

where $K = \frac{\cos \phi}{\cos \beta \cos \alpha}$

$\phi \rightarrow$ phase angle between current and voltage

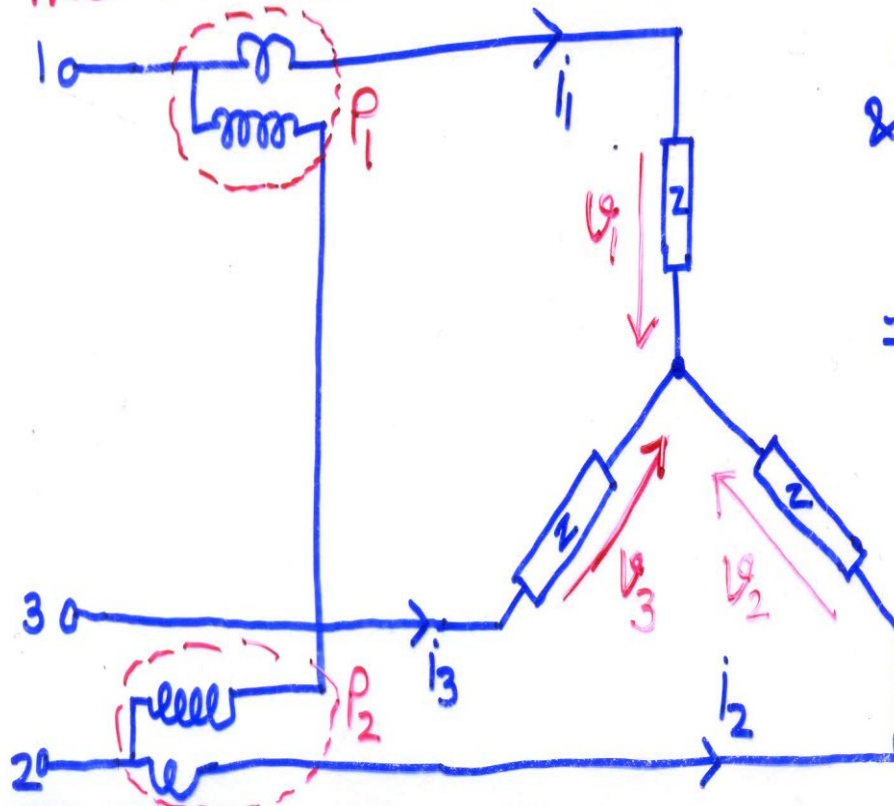
$\beta \rightarrow$ angle by which I_p lags V_s on account of inductance of pressure coil

$\alpha \rightarrow$ phase angles between currents in C.C. & P.C. of wattmeter

$W \rightarrow$ Wattmeter reading

Two Wattmeter Method

TWO WATTMETER METHOD



$$p = V_1 i_1 + V_2 i_2 + V_3 i_3$$

$$\& i_1 + i_2 + i_3 = 0$$

$$i_3 = -(i_1 + i_2)$$

$$\Rightarrow p = V_1 i_1 + V_2 i_2 + V_3 [-(i_1 + i_2)]$$

$$= i_1 (V_1 - V_3) + i_2 (V_2 - V_3)$$

$$= P_1 + P_2$$

(28.1)